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Abbreviations

AAAI: Association for the Advancement of Artificial Intelligence

ACL: Association for Computational Linguistics

AI: Artificial Intelligence

API: Application Programming Interface

CNN: Convolutional Neural Network

CRNN: Convolutional Recurrent Neural Network

CV: Computer Vision

CVPR: Conference on Computer Vision and Pattern Recognition

CVPRW: Conference on Computer Vision and Pattern Recognition Workshops

DPNet: Dual-Path Network

ECCV: European Conference on Computer Vision

ICCV: International Conference on Computer Vision

ICDAR: International Conference on Document Analysis and Recognition

iOS: iPhone Operating System

NLP: Natural Language Processing

NPR: Nepalese Rupee

OCR: Optical Character Recognition

OHRBench; OCR under Real-World Noise Benchmark

TrOCR: Transformer-based Optical Character Recognition

UI: User Interface

Chapter 1

Introduction

In this advanced era, the use of technology is increasing day by day. For every sector including language translators various tools are used. Various signboards can be seen in different areas such as streets, government offices, stations, shops, restaurants, and tourist spots. These signs may contain information that is difficult to understand for foreigners who do not know the local language.

The concept of the proposed software product is known as AI-based Signboard Reader and Translator. It involves developing a software product that would allow for quick and reliable detection, recognition, and translation of information provided on signboards. In other words, one can describe this software product as utilizing OCR and Computer Vision technologies together with AI-powered language translation to convert images captured through the camera to text content in a particular language.

The primary aim of this software product is to offer a way for foreigners or people living in multinational environments to access the information provided on signboards in various locations. Using this technology, customers would be able to take pictures of signboards using their smartphones' cameras, and immediately see translations of text content.

This project demonstrates the practical application of Artificial Intelligence in solving real-world communication challenges and contributes to the development of intelligent technologies that make our real world works more easy and faster.

1.1 Background

Artificial Intelligence (AI) is being used in many areas of daily life to make everyday tasks easier and faster. Technologies such as Optical Character Recognition (OCR), Computer Vision, and Language Translation have improved the way people access and understand information.

In the past, tourists needed a guide for almost everything, including signboard translation, and hiring a guide was expensive. However, nowadays, people travel, study, work, and visit different places more than ever before. During these activities, they often come across signboards, notices, and information boards written in languages they do not understand. These signboards may be found on roads, hospitals, government offices, public transport stations, shops, restaurants, and tourist areas.

Understanding the information on such signboards is important, but language differences can make it difficult. Recent developments in OCR and translation technologies make it possible to detect text from images and translate it automatically. By combining

these technologies, a system can help users understand signboard information quickly and easily.

1.2 Motivation

The main motivation for this project is to help people overcome language barriers. Travelers, tourists, students, and foreign visitors often face problems when they cannot understand signboards written in a different language.

Although translation applications are available, many of them require users to enter text manually, which can be slow and inconvenient. An AI-based Signboard Reader and Translator can make this process easier by detecting text directly from images and translating it automatically. This project also provides an opportunity to apply AI, OCR, and translation technologies to solve a real-world problem and make information more accessible to everyone.

1.3 Problem Statement

Today, the world is highly connected; thus, it is common for people to travel, study, or work in areas where multiple languages exist, such as English, Nepali, Chinese, and Spanish.

Information found on signs along roads, public transport, government buildings, hospitals, universities, tourism sites, shopping malls, and eateries may sometimes be difficult to understand by those who do not know the language of that particular region. In various parts of Nepal, signboards in hospitals, on roads, and in public transportation may be written in local languages such as Tamang and Tharu.

While there are available translation apps, the process of translation through those apps involves inserting text by the user; therefore, it takes too much time. Moreover, in Nepal and many other countries, multiple languages are used, as each country has multiple castes, traditions, and cultures.

Therefore, there is a need for a smart solution that can detect text on signboards and translate it into the user's preferred language. An intelligent device that could recognize text in signboards and translate them for the user would enable the latter to obtain useful information immediately without knowing the local language.

Thus, the proposal for an intelligent system called AI-based Signboard Reader and Translator that could integrate computer vision, and artificial intelligence-based technologies of translation would solve the issue raised above.

1.4 Objectives

To develop an AI-based system that detects, reads, and translates signboard text into a user-understandable language using OCR.

1.4.1 Specific Objectives

- To capture clear images of public signboards using standard smartphone cameras or digital cameras. This process helps the app get visual data from different outdoor

places like roads, stations, and hospitals, making sure that the text is easy for the system to see and read under normal daytime conditions.

- To detect and isolate text lines from complex backgrounds using computer vision tools and OCR software. This step ensures that the system can separate words and characters from the background of the signboards, even when there are challenges like uneven outdoor lighting, different fonts, or slightly blurry images.
- To identify multiple local and international languages automatically before translating them into the language that the user prefers. This includes adding support for regional languages and local dialects like Tamang and Tharu, so that foreign tourists and local travelers can easily understand signs found in different parts of the country.
- To translate the recognized signboard text quickly and accurately by using artificial intelligence translation methods. The main focus is to provide correct translations in a short amount of time, allowing users to get important local information instantly without needing to type out any words manually on their phones.

1.5 Significance of the Study

This project is important because it helps people overcome language barriers when reading signboards and public information. Many travelers, tourists, international students, and foreign workers visit places where they may not understand the local language. In such situations, important information on roads, hospitals, transportation stations, government offices, shops, and tourist sites can be difficult to understand. The proposed system can help users access this information more easily by providing automatic text recognition and translation.

The project is also significant because it demonstrates the practical use of modern technologies such as AI, OCR, Computer Vision, and Machine Translation. By combining these technologies into a single system, the project provides a smart and efficient solution for real-world communication problems.

In addition, the proposed system can save time and reduce the effort required to manually enter text into translation applications. Users will be able to capture an image of a signboard and receive the translated information quickly. This can improve convenience and accessibility, especially in multilingual environments.

Furthermore, this study contributes to the development of intelligent applications that support communication between people from different linguistic backgrounds. It also provides valuable learning and research opportunities in the fields of Artificial Intelligence, Computer Vision, and Natural Language Processing, which are important areas of modern technology.

Chapter 2

Literature review

2.1 Theoretical Background

The proposed system combines three main technologies to read and translate signboards automatically. These technologies work together in a sequence to convert an image into translated text.

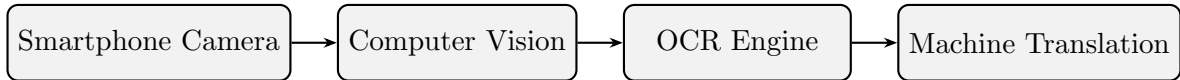


Figure 2.1: Core processing pipeline

- **Computer Vision** Computer Vision enables computers to interpret and understand digital images. In this project, it is used to scan the entire image of a signboard and find exactly where the text is located. It helps the system separate the text from complex background elements like trees, buildings, or poles. The system uses simple machine learning models to detect text boundaries even when the lighting is poor or the image is taken at an angle.
- **Optical Character Recognition (OCR)** Optical Character Recognition is a technology that turns text inside an image into editable digital text [8]. Once Computer Vision finds the text blocks, the OCR engine reads the shapes of the letters and characters [4]. It then converts those shapes into actual digital words that a computer can process. This project uses standard OCR tools to quickly recognize different fonts and handwriting styles on signs [8].
- **Machine Translation** Machine Translation is a branch of artificial intelligence that automatically translates text from one language to another. This system uses modern translation models to convert the recognized text into the user's preferred language. Instead of translating word for word, the system looks at the whole phrase to provide natural and meaningful translations. This is very helpful for translating local names and signs instantly [6].

2.2 Related Works

Automatic text recognition has been an active research area for many years. Early Optical Character Recognition (OCR) systems were mainly developed to recognize printed

text from scanned documents and books. One of the most widely used OCR engines is Tesseract OCR, which converts printed characters into machine-readable text with high accuracy under controlled conditions. Although Tesseract performs well on clean images with simple backgrounds, its performance decreases when text is captured from real-world scenes where images contain uneven lighting, shadows, blur, or complex backgrounds [9].

To overcome these challenges, researchers shifted their focus from traditional OCR to scene text recognition. Instead of recognizing text only from scanned documents, scene text recognition aims to detect and read text from natural images such as road signs, shop names, advertisements, and public information boards. Convolutional Recurrent Neural Networks (CRNN) introduced an effective approach by combining convolutional neural networks for feature extraction with recurrent neural networks for sequence prediction, significantly improving text recognition from natural scene images [3].

With the rapid advancement of deep learning, more powerful models have been proposed to further improve scene text recognition. DPNet combines Convolutional Neural Networks (CNN) with Transformer architectures to capture both local image features and long-range contextual information, leading to more accurate recognition of complex scene text [2]. Similarly, TrOCR utilizes a Transformer-based architecture with pre-trained models to improve OCR performance on challenging images containing different fonts, distorted text, and low-quality visual conditions [8]. These modern approaches have demonstrated significant improvements over traditional OCR systems.

Besides text recognition, machine translation has also experienced remarkable progress. Recent multilingual Neural Machine Translation (NMT) models use deep learning techniques to understand the contextual meaning of complete sentences rather than translating individual words. This results in more accurate and natural translations across multiple languages, making communication easier for users from different linguistic backgrounds [6]. The availability of such multilingual translation models has expanded the practical applications of OCR-based systems.

The integration of OCR and machine translation technologies has enabled the development of practical applications that assist users in understanding foreign-language text. Google Cloud Translation API provides automatic language detection and translation services that can be integrated with OCR systems to translate text extracted from images in real time [7]. By combining image processing, text recognition, and language translation, these technologies have become useful for applications such as document translation, travel assistance, and multilingual information access.

Although significant progress has been achieved, several challenges still remain. According to recent studies on OCR under real-world noisy conditions, the accuracy of existing systems decreases when images contain blur, shadows, reflections, occlusion, complex backgrounds, or unusual viewing angles [10]. Furthermore, most existing systems mainly focus on widely spoken international languages and provide limited support for regional or low-resource languages. These limitations reduce the effectiveness of current applications when used in multilingual countries where local languages frequently appear on public signboards.

Based on these previous studies, the proposed AI-based Signboard Reader and Translator aims to integrate computer vision, OCR, language identification, and machine translation into a single mobile application. The proposed system builds upon the strengths of recent scene text recognition and multilingual translation techniques while addressing their limitations by focusing on real-world outdoor signboards and improving support for regional languages. The application is expected to provide fast, accurate, and user-friendly translation of multilingual signboards, making information more accessible to

travelers, tourists, students, and local users.

2.3 Research Gap Analysis

Although there are many translation apps available today, they still have major limitations when used in real-world scenarios. Based on our review of existing systems, we found these main research gaps:

- **Lack of Support for Local Languages:**

Most current translation tools focus only on major global languages like English, Spanish, or Chinese. They do not support regional languages or local dialects, such as Tamang or Tharu, which are widely used on signs in countries like Nepal.

- **Poor Performance in Challenging Environments:**

Existing apps work well with clean images but fail when images are taken outdoors. They struggle to read text when there are complex backgrounds, uneven shadows, bad lighting, or artistic fonts on a signboard[10].

In many existing tools, the text reader (OCR) and the translator work as separate pieces. This causes slow processing times and creates errors when sending the text from the reader to the translation engine.

Our proposed application, AI-based Signboard Reader and Translator, solves these problems by combining a smart text detector and a fast translation engine into one single app. It will automatically find and read text without requiring the user to type anything. Most importantly, it will focus on adding better support for local languages and handling blurry or poorly lit outdoor images much better than standard apps

Chapter 3

Proposed Methodology

3.1 Proposed Project Approach

This project will follow the Agile Software Development Methodology. This approach means we will build the application in small, separate parts instead of trying to finish everything at once. This method helps our team test every feature as soon as it is made, catch errors early, and easily make changes based on feedback.

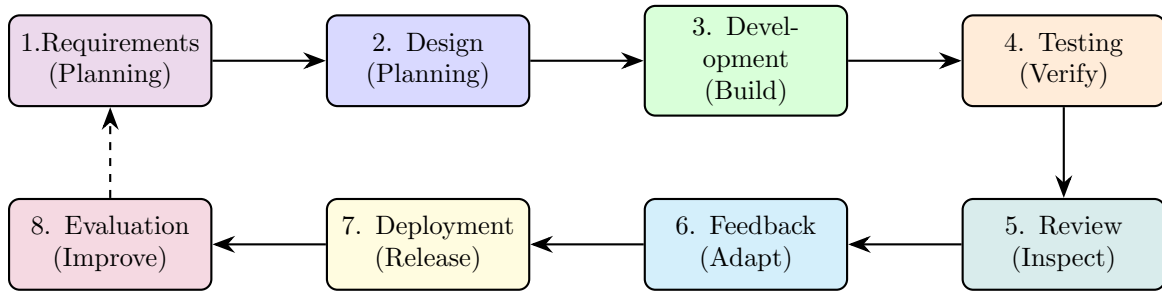


Figure 3.1: Agile Software Development Methodology used in the proposed project

This project follows the Agile Software Development Methodology, which focuses on developing the system through iterative and incremental cycles called sprints. Instead of completing the entire application at once, the project is divided into smaller tasks that are designed, developed, tested, and improved continuously based on feedback. This approach enables the team to identify problems early, incorporate suggestions from the supervisor, and deliver a better-quality application.

The development process begins with the planning phase, where the project requirements are gathered. During this stage, the team identifies the languages that the application should support, such as English, Nepali, Tamang, and Tharu, and determines the image conditions that the system should handle, including outdoor lighting, image blur, and different viewing angles. Based on these requirements, the project backlog is prepared and the features are prioritized.

After planning, the design phase is carried out. In this phase, the overall architecture of the system is designed by defining how the Computer Vision module, Tesseract OCR engine, language identification module, and Google Cloud Translation API will interact with one another. The development tasks are then divided into manageable sprints so that each feature can be implemented and tested independently.

The next stage is development, where the actual implementation of the application takes place. The team develops the image capture module, performs image pre-processing

using OpenCV, integrates Tesseract OCR for text recognition, and connects the Google Cloud Translation API to translate the extracted text into the user's preferred language. Each module is implemented gradually according to the sprint plan.

Once a module is completed, it enters the testing phase. Individual components are verified to ensure they function correctly before integration. For example, the OCR module is tested to determine whether it can accurately recognize text from signboards captured under different lighting conditions, with varying fonts, or from slightly blurred and angled images. Similarly, the translation module is tested to evaluate the accuracy of the translated output.

Following testing, the completed work is presented during the review phase, where the development team and project supervisor evaluate whether the implemented features satisfy the project objectives. Suggestions and observations collected during the review are then incorporated into the feedback phase, allowing the team to improve the application in the next sprint. For instance, if the OCR engine struggles to recognize regional fonts or the translation engine produces inaccurate translations for local languages, the necessary improvements are planned for the following iteration.

After the improved version is ready, the application is deployed for trial use so that users such as tourists, travelers, and students can test it in real-world environments. Finally, the project enters the evaluation and continuous improvement phase, where user feedback and testing results are analyzed to enhance the application's recognition accuracy, translation quality, processing speed, and support for additional regional languages. This iterative cycle continues until the application satisfies all the project requirements.

3.2 Tools and Technologies

To make these steps work, the project will use the following simple and reliable development tools:

- Python:

Python is the primary programming language used to develop the application's backend logic. It is used for image processing, OCR integration, language identification, and communication between different system modules.

- OpenCV:

OpenCV is an open-source computer vision library used for image pre-processing tasks such as resizing, noise removal, contrast enhancement, and text region detection before OCR is performed.

- Tesseract OCR:

Tesseract OCR is an open-source Optical Character Recognition engine used to convert text present in signboard images into machine-readable digital text. In this project, it works together with OpenCV after image pre-processing to recognize characters accurately before passing the extracted text to the language identification and translation modules. Tesseract was selected because it is free, reliable, supports multiple languages, and integrates easily with Python applications[9].

- Google Translation API:

OCR Process Flow

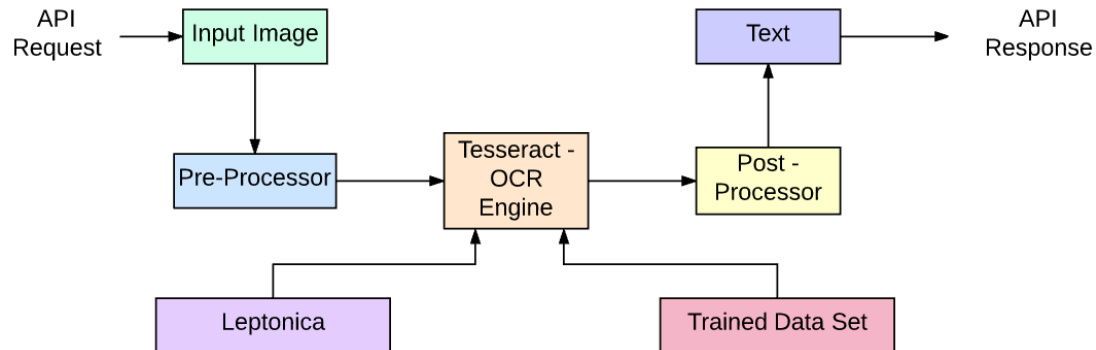


Figure 3.2: Process of OCR

Google Cloud Translation API is used to automatically detect the source language and translate the extracted text into the user's preferred language. It supports multilingual translation with high accuracy[7].

- Flutter / React Native:

Flutter or React Native will be used to develop the mobile application's user interface. These frameworks provide a responsive, user-friendly interface for both Android and iOS devices.

3.3 System Steps and Tools

This section explains the exact step-by-step process the application follows to translate a signboard, along with the technology tools used to build it. Steps Involved:

- Upload Image:

The process begins when the user captures a live image of a signboard using the smartphone camera or selects an existing image from the device gallery. The image serves as the input to the system and may contain signboards from roads, hospitals, public offices, shops, restaurants, or tourist locations. High-quality images improve the overall recognition accuracy.

- Pre-process Image:

The captured image is pre-processed using OpenCV to improve its quality before text recognition. The system converts the image into grayscale, removes unwanted noise, adjusts brightness and contrast, and applies thresholding techniques where necessary. These operations help reduce the effects of shadows, poor lighting, and image blur, making the text easier to detect and recognize.

- Detect Text Using Computer Vision Techniques:

After image enhancement, computer vision techniques are used to detect the regions containing text. The system identifies the boundaries of words and characters while ignoring unnecessary background objects such as buildings, vehicles, trees, or advertisements. Only the detected text regions are passed to the OCR engine for further processing.

- Text Recognition using OCR:

The detected text regions are processed using the Tesseract OCR engine. Tesseract analyzes the shapes of individual characters, groups them into words and sentences, and converts the image-based text into machine-readable digital text. This extracted text becomes the input for the language identification and translation stages. Tesseract was selected because it is an open-source OCR engine that provides reliable recognition for printed text and integrates efficiently with Python applications [9].

- Language Identification:

Once the text has been extracted, the system automatically determines the language in which the signboard is written. Identifying the correct source language is important because it allows the translation engine to apply the appropriate translation model and improve translation accuracy.

- Machine Translation:

After identifying the source language, the recognized text is translated into the user's preferred language using Google Cloud Translation API. The translation engine processes complete words and sentences to generate meaningful translations rather than translating individual words. This enables users to understand the information displayed on the signboard quickly and accurately [7].

- Produce Output:

Finally, the translated text is displayed clearly on the mobile application's user interface. The user can read the translated content immediately after processing. In future versions of the project, additional features such as text-to-speech, saving translation history, and offline translation support may also be incorporated to improve usability.

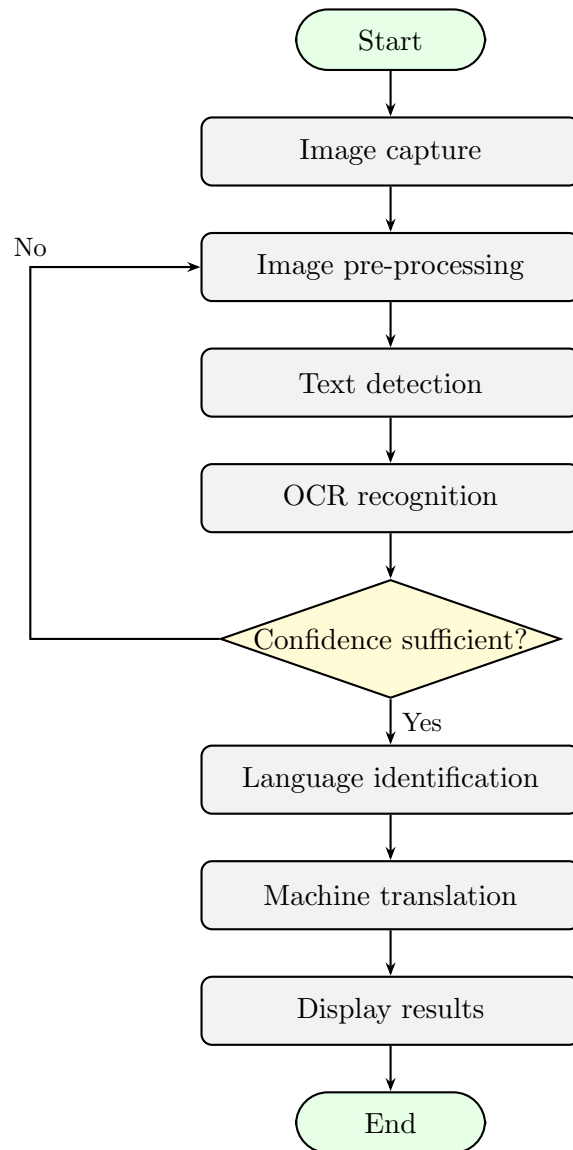


Figure 3.3: System flowchart for AI-based Signboard Reader and Translator

Chapter 4

Expected Outcomes

4.1 Expected Results

The project aims to achieve the following successful results upon completion:

- **Functional Signboard Translation Application:**
A working mobile application that allows users to scan and read real-world signboards instantly.
- **Automatic extraction of text from images:**
The system will pull words out of raw pictures automatically without requiring the user to type anything by hand.
- **Detection of source language:**
The app will automatically recognize which language is written on the sign before starting the translation process.
- **Multilingual accurate translation:**
The system will provide correct translations for multiple local and international languages so travelers can understand regional terms.
- **User-friendly interface:**
A simple, clean, and uncomplicated screen design that anyone can navigate easily without technical training.

4.2 Validation Strategy

We will use the following methods to confirm that our project meets all the required quality standards:

- **User Acceptance Testing:**
Sharing the app with real travelers, students, and tourists to see if it actually helps them solve their real-world language problems.
- **Expert Review:**
Showing the complete system to college teachers and IT experts to get professional feedback on our code and model choices.

- **Accuracy Testing:**
Evaluating the system using quantitative performance metrics to measure exactly how well the app detects text boundaries and how correct the final translations are.
- **Usability Testing:**
Checking if the buttons are easy to find, the layout is clear, and regular users can operate the app without getting confused.

4.3 Testing Strategy

To find and fix software bugs early, we will use these four standard testing steps:

- **Unit Testing:**
Testing every small piece of code on its own, such as checking just the image brightness filter to make sure it works perfectly.
- **Integration Testing:**
Testing how well different parts of the app talk to each other, like making sure the text reader sends the correct words straight to the translation engine.
- **System Testing:**
Testing the entire mobile application from start to finish on different Android and iOS phones to guarantee the whole system runs smoothly.
- **Performance Testing:**
Measuring how fast the application loads, ensuring that translation occurs within a few seconds, and verifying that it does not drain too much phone battery.

Chapter 5

Time Schedule and Budget

5.1 Timeline

Task	May 2026		Jun 2026				Jul 2026				Aug 2026		
	May 24	May 31	Jun 07	Jun 14	Jun 21	Jun 28	Jul 05	Jul 12	Jul 19	Jul 26	Aug 02	Aug 09	Aug 16
Literature Review													
Requirement Analys													
System Design													
Development													
Testing													
Documentation													
Final Submission													

Table 5.1: Project Timeline

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